Companion to Bird & de Moor, Algebra of Programming

§1. Programs

§1.1. Datatypes

§1.2. Natural numbers

§1.3. Lists

§1.4. Trees

§1.5. Inverses

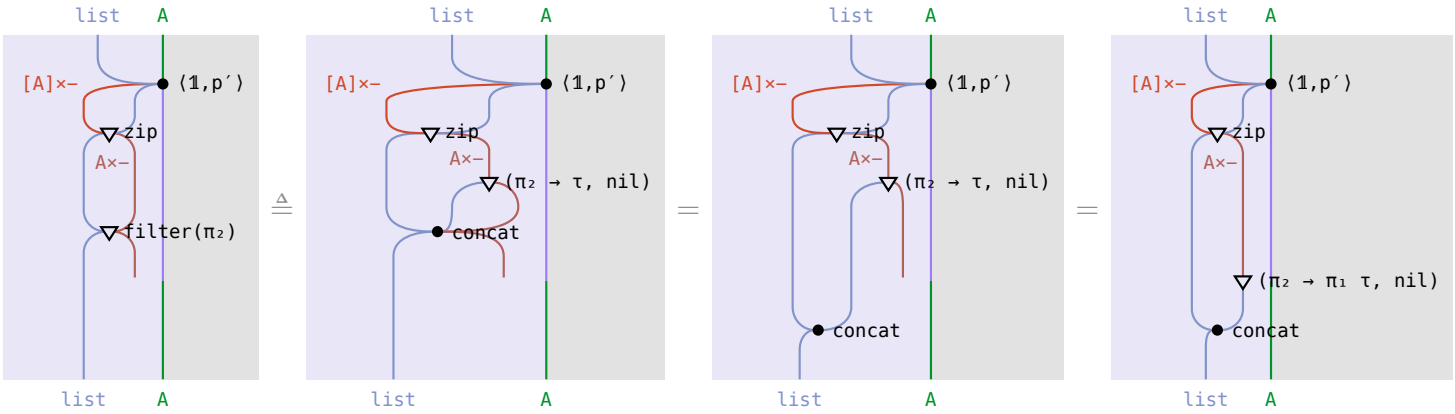
§1.6. Polymorphic functions

§1.7. Pointwise and point-free

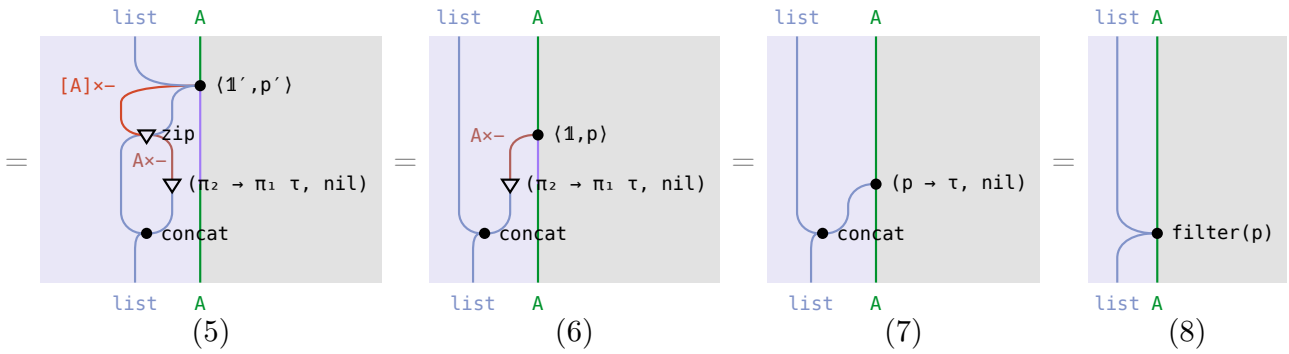
`⟨𝟙,p′⟩ zip filter(π₂) π₁′ = filter(p)` (1.7a)

pairing each element with its answer under `p` (`⟨𝟙,p′⟩ zip`), keeping the pairs answered `true` (`filter(π₂)`) and dropping the answers (`π₁′`) is `filter(p)`

the starting composite



1	`⟨𝟙,p′⟩ zip filter(π₂) π₁′`	definition of filter: `filter(p) ≜ (p → τ, nil)′ concat`
2	`≜ ⟨𝟙,p′⟩ zip (π₂ → τ, nil)′ concat π₁′`	(1.6): `concat f′ = f′′ concat` (1.8) backwards: `f′ g′ = (fg)′`
3	`= ⟨𝟙,p′⟩ zip (π₂ → τ, nil) π₁′′ concat`	(1.11): `(p → f, g)h = (p → fh, gh)`; (1.5): `τ f′ = f τ`; (1.4): `nil f′ = nil`
4	`= ⟨𝟙,p′⟩ zip (π₂ → π₁ τ, nil)′ concat`	(1.9) backwards: `𝟙 = 𝟙′`



5	`= ⟨𝟙′,p′⟩ zip (π₂ → π₁ τ, nil)′ concat`	(1.7): `⟨f′,g′⟩ zip = ⟨f,g⟩′` (1.8) backwards: `f′ g′ = (fg)′`
6	`= ⟨𝟙,p⟩ (π₂ → π₁ τ, nil)′ concat`	(1.10): `h(p → f, g) = (hp → hf, hg)`; (1.1): `⟨f,g⟩π₁ = f`; (1.3): `f nil = nil` (1.12): `𝟙 f = f`
7	`= (p → τ, nil)′ concat`	definition of filter
8	`= filter(p)`	